Xifeng Zhu*, Dinghui Zou, Yanhua Huang, Junming Cao, Yuping Sun, Bing Chen and Xiaoying Chen

Physiological responses of *Porphyra haitanensis* (Rhodophyta) to copper and cadmium exposure

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Abstract: The aim of the present work was to evaluate the effects of copper (Cu) and cadmium (Cd) on the physiology of the red macroalga *Porphyra haitanensis* Chang and Zheng. The alga was cultured in media with different Cu or Cd concentrations for 7 days. The relative growth rate was significantly reduced at Cu concentrations of 0.10 mg l^{-1} and above or at Cd concentrations of 4 mg l^{-1} and above. Chlorophyll *a* (Chl *a*), carotenoids (Car), phycoerythrin (PE) and phycocyanin (PC) contents decreased at high Cu or Cd concentrations, and PE content increased at 0.10 mg l^{-1} Cu or 4–8 mg l^{-1} Cd. Moreover, the optimal quantum yield, maximum net photosynthetic rate (P_m) and apparent photosynthetic efficiency (α) were significantly reduced at high Cu or Cd concentrations. In contrast, malondialdehyde (MDA) content and superoxide dismutase (SOD) activity increased in response to Cu or Cd treatment. Catalase activity increased at 2–8 mg l^{-1} Cd and significantly declined at 0.50 mg l^{-1} Cu or 10 mg l^{-1} Cd. Nitrate reductase (NR) activity declined at high Cu or Cd concentrations. Consequently, both Cu and Cd were physiological stressors for *P. haitanensis*, with Cu being more toxic than Cd. This work has generated valuable knowledge about the toxic effects of Cu and Cd on macroalgae.

***Corresponding author: Xifeng Zhu,** The Key Laboratory of Animal Nutrition and Feed Science (South China) of Ministry of Agriculture, Guangdong Open Laboratory of Animal Breeding and Nutrition, Guangdong Key Laboratory of Animal Breeding and Nutrition, Institute of Animal Science, Guangdong Academy of Agricultural Sciences, Guangzhou, Guangdong 510640, China, e-mail: xfzhu1983@163.com
Dinghui Zou: College of Environment and Energy, South China University of Technology, Guangzhou 510006, China; and The Key Laboratory of Pollution Control and Ecosystem Restoration in Industry Clusters, Ministry of Education, China
Yanhua Huang, Junming Cao, Yuping Sun, Bing Chen and Xiaoying Chen: The Key Laboratory of Animal Nutrition and Feed Science (South China) of Ministry of Agriculture, Guangdong Open Laboratory of Animal Breeding and Nutrition, Guangdong Key Laboratory of Animal Breeding and Nutrition, Institute of Animal Science, Guangdong Academy of Agricultural Sciences, Guangzhou, Guangdong 510640, China

Keywords: cadmium; copper; growth; photosynthesis; *Porphyra haitanensis*.

Introduction

Marine macroalgae are major primary producers in many marine ecosystems and play an important role in food chains in coastal regions (Ramus 1992, Zhu et al. 2015). Coastal waters are most affected by anthropogenic pollutants such as metals. Macroalgae can accumulate metals dissolved in water and transfer them to higher trophic levels (Torres et al. 2008). Copper (Cu) and cadmium (Cd) are two major metal pollutants arising from various anthropogenic sources, such as industrial discharges, mining drainage and pesticides (Millward and Turner 2001, Brown and Newman 2003, Zhu et al. 2011). Many papers have reported that macroalgae could be heavily impacted by Cu and Cd in marine ecosystems (Collén et al. 2003, Huang et al. 2010, Zhu et al. 2011). Researchers are now paying greater attention to the toxic effects of Cu and Cd on algae. Cu is an essential micronutrient for algae and is required for several important metabolic and physiological processes (Brown and Newman 2003). However, Cu is generally considered to be one of the most toxic metals at elevated concentrations (Gledhill et al. 1997, Han et al. 2008, Hurd et al. 2014). The redox-active nature of Cu can induce severe phytotoxic symptoms at higher levels in algae, such as inhibition of growth and photosynthesis, reduction in chlorophyll content, generation of reactive oxygen species (ROS), increase of malondialdehyde (MDA) content and antioxidant enzyme activities (Brown and Newman 2003, Collén et al. 2003, Han et al. 2008, Zhu et al. 2011). Cd is a non-essential and generally toxic metal for algae (Fernandez-Leborans and Novillo 1996). The toxic effects of Cd have been reported in *Gracilaria lemaneiformis* (Bory) Greville and *Hizikia fusiformis* (Harvey) Okamura (Xia et al. 2004, Zhu et al. 2011). It was shown that Cd could inhibit the growth rate and the photosynthetic electron transport chain, decrease the content of chlorophyll, change antioxidant enzyme activities and cause lipid peroxidation in these macroalgae (Xia et al. 2004, Zhu et al. 2011).

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Recent studies have shown that macroalgae have a large capacity to take up inorganic nutrients and that they can be used as biomonitors of metal contamination because of their high potential for metal accumulation (Yang et al. 2006, Abdallah and Abdallah 2008, Huang et al. 2010, Gaete Olivares et al. 2016). As marine metal pollutants, especially Cu and Cd, have become more serious in coastal regions in recent years, we should pay more attention to the toxic effects of Cu and Cd on marine macroalgae (Zhu et al. 2011, Gaete Olivares et al. 2016). Moreover, the protective mechanisms that enable macroalgae to resist the toxic effects of metals should be researched. Several studies have shown that antioxidants and antioxidant enzymes, such as carotenoids (Car), superoxide dismutase (SOD) and catalase (CAT) scavenge ROS and thus protect macroalgae (Collén et al. 2003, Zhu et al. 2011). In addition, nitrate reductase (NR) is responsible for reducing nitrate to nitrite in algae (Young et al. 2005). Our previous work found that NR was sensitive to metals and could be significantly affected by Cu and Cd (Zhu et al. 2011). Metal accumulation varied between macroalgae and different species had differential tolerance to metals (Baumann et al. 2009). It will be of value to know more about the physiological responses of these species to metal stress.

Porphyra haitanensis Chang and Zheng is an important seaweed used as sea-vegetable food in China, being cultivated on a commercial scale in southern China. *Porphyra haitanensis* is also important as a source for biochemicals and pharmaceuticals (Li et al. 2010). In recent years, however, its natural abundance has decreased with the increasing level of marine pollution, including pollution by metals and pesticides, in Chinese coastal regions, such as Nan'ao Island, which is an important area for algal cultivation and biodiversity (Zhu et al. 2015). The physiological characteristics of *P. haitanensis* have been well studied (Tang and Fei 1997, Zou and Gao 2002, Jiang et al. 2009), but the physiological responses of this alga to metals have not been documented much. *Porphyra haitanensis* is a representative red macroalga of the intertidal zone that can be used as a model for the study of Cu and Cd tolerance. Moreover, we can compare the toxic effects of the metals on *P. haitanensis*.

The aim of this study was to use a series of Cu or Cd concentrations, similar to and higher than those present in Nan'ao Island where *P. haitanensis* is cultivated, to assess the toxic effects of Cu or Cd on the growth, photosynthesis, chlorophyll and phycobiliprotein content, lipid peroxidation, SOD activity, CAT activity and NR activity in this species. The results will be used to evaluate the physiological response mechanisms of *P. haitanensis* to Cu or Cd exposure and to help predict the toxic effects of Cu or Cd on this macroalga in the future.

Materials and methods

Plant material

Thalli of *Porphyra haitanensis* (3–5 cm in length) were collected from Yun'ao Bay, Nan'ao Island, Shantou, China (23°20′ N, 116°55′ E), during December 2009. This region has no metal exposure history and the concentrations of Cu or Cd are very low. The algae collected were carefully washed and transported to the laboratory in plastic buckets within 2 h. Samples were maintained in a glass aquarium tank with natural seawater (Cu concentration < 0.10 μg l^{-1}; Cd concentration < 0.02 μg l^{-1}) at constant temperature (20°C) and irradiance (100 μmol m^{-2} s^{-1}) with a 12-h photoperiod. After 2 days of laboratory maintenance, healthy individuals were selected for use in subsequent experiments.

Experimental design

Thalli of about 2 g fresh weight (FW) of *Porphyra haitanensis* (3–5 cm in length) were cultured in Erlenmeyer flasks that contained 2 l of natural seawater enriched with 100 μM $NaNO_3$ and 10 μM $Na_2HPO_4 \cdot 12H_2O$. Metals were added as solutions of $CuSO_4 \cdot 5H_2O$ and $CdCl_2 \cdot 2.5H_2O$. The initial concentrations were 0, 0.025, 0.05, 0.10, 0.25 and 0.50 mg l^{-1} for Cu and 0, 2, 4, 6, 8 and 10 mg l^{-1} for Cd. Three replicates were carried out for each set of treatments. The culture media were renewed every day. Treated algae were cultured in an illuminated incubator (GXZ-300D, Jiangnan Instrument Factory, Ningbo, China). The light and temperature conditions were as mentioned above and the 12:2-h light cycle was set from 07:00 to 19:00 (local time). After 7 days of exposure to Cu and Cd, the thalli were harvested and physiological indicators were determined.

Growth rate

The FW of algae was measured at the end of incubation to estimate growth. Relative growth rate (RGR) was calculated according to the equation: RGR (% day^{-1}) = ln (W_t/W_0)/t*100, where W_0 = initial FW, and W_t = FW after t days.

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Pigment estimation

The contents of chlorophyll *a* (Chl *a*) and total Car were determined using an ultraviolet spectrophotometer (UV-2450, Shimadzu, Kyoto, Japan). About 0.2 g of FW algae were ground in 10 ml of 100% methanol and extracted at 4°C in darkness for 24 h. Chl *a* and Car concentrations were estimated according to Wellburn (1994).

Phycoerythrin (PE) and phycocyanin (PC) contents were estimated according to the method by Beer and Eshel (1985). Approximately 0.2 g of FW algae were placed in 5 ml of 0.1 M phosphate buffer (pH 6.8), ground using a mortar and pestle (with a little acid-washed sand), rinsed with a further 5 ml of buffer and then centrifuged at 1000 × *g* for 10 min. Phycobiliprotein contents in the supernatant were calculated using the following formulae:

$$PE = [(A_{564} - A_{592}) - (A_{455} - A_{592}) \times 0.20] \times 0.12$$

and

$$PC = [(A_{618} - A_{645}) - (A_{595} - A_{645}) \times 0.51] \times 0.15,$$

where A is the absorbance at each wavelength.

Chlorophyll fluorescence and photosynthetic oxygen evolution

Chlorophyll fluorescence was measured using a pulse-amplitude-modulated (PAM) fluorometer (PAM-WATER-EDF, Walz, Effeltrich, Germany). Algae were initially dark-incubated before being measured. The optimal quantum yield (F_v/F_m) of photosystem (PS) II was determined using the equation: $F_v/F_m = (F_m - F_o)/F_m$ (Maxwell and Johnson 2000). Here, F_m and F_o are the maximum and minimum fluorescence of the dark-adapted stage of PS II and F_v is the difference between F_m and F_o.

Photosynthetic O_2 evolution rates at different irradiances (P vs. I curves) were measured using a Clark-type oxygen electrode (YSI Model 5300, Yellow Springs Instrument Co., Yellow Springs, OH, USA) at 20°C. Illumination was provided by a halogen lamp and irradiance was measured with a quantum sensor (SKP 200, ELE International, Leighton Buzzard, UK) at the surface of the chamber. Irradiance was varied from 0 to 700 μmol m^{-2} s^{-1} by altering the distance between the electrode chamber and the light source. Approximately, 0.2 g of FW algae were placed in the electrode chamber containing 10 ml of filtered natural seawater, which was magnetically stirred. The maximum net photosynthetic rate (P_m), apparent photosynthetic efficiency (α) and dark respiration rate (R_d) were calculated by a non-linear curve fit of the P-I curve (Henley 1993).

MDA determination

The content of MDA was determined by using the thiobarbituric acid (TBA) reaction (Health and Packer 1968). About 0.2 g of FW algae was homogenized in 5 ml of 0.1% trichloroacetic acid (TCA). The homogenate was centrifuged at 10,000 × *g* for 5 min at 4°C. For every 1 ml of supernatant, 4 ml of 20% TCA containing 0.5% TBA was added. After heating at 95°C for 30 min, the mixture was cooled in an ice bath and centrifuged at 10,000 × *g* for 15 min at 4°C. The absorbance of the supernatant at 532 nm was recorded and corrected for non-specific turbidity by subtracting the value at 600 nm. The concentration of MDA was calculated using the extinction coefficient of 155 mM^{-1} cm^{-1}.

Activities of antioxidant enzymes and NR

SOD (EC 1.15.1.1) activity was determined by measuring the inhibition of nitro blue tetrazolium (NBT) reduction at 560 nm (Beauchamp and Fridovich 1971). About 0.2 g of FW algae were ground in liquid N_2 and extracted in 1 ml of 50 mM potassium phosphate buffer (pH 7.8) containing 0.1 mM ethylenediaminetetraacetic acid (EDTA), 0.3% (v/v), Trion X-100 and 4% (w/v) polyvinyl pyrrolidone (pvpp). The extraction solution was filtered through cheese cloth and centrifuged at 10,500 × *g* for 20 min at 4°C. The reaction mixture contained 3 ml of 13 mM DL-methionine, 75 μM NBT, 2 μM riboflavin, 0.1 mM EDTA and 1 ml of enzyme extract. Before reading the absorbance at 560 nm, the test tubes were shaken and placed 30 cm below a 15-W fluorescent lamp for 10 min at 25°C. One unit of SOD (per mg protein) was defined as the amount causing 50% inhibition of the photochemical reduction of NBT. Protein content was determined spectrophotometrically at 595 nm by the method of Bradford (1976) using bovine serum albumin (BSA, Sigma) as a standard.

Catalase (EC 1.11.1.6) activity was determined using a ferrithiocyanate method adapted from Aebi (1984) and Cohen et al. (1996). Approximately 0.2 g of FW algae were ground in liquid N_2 and extracted with 5 ml of 10 mM potassium phosphate buffer (pH 7.0). The extraction solution was centrifuged at 10,500 × *g* for 20 min at 48°C. The supernatant was used to measure CAT activity and protein content. Ferrous ions (10 mM) and thiocyanate (2.5 M) were used for the spectrophotometric determination of hydrogen peroxide (H_2O_2). Ferrithiocyanate was measured in the spectrophotometer and the stable visible absorption was measured at 492 nm at 25°C. Protein content was determined by the same method as previously described.

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NR (EC 1.7.1.1) activity was measured following the *in situ* method of Corzo and Niell (1991). About 0.2 g of FW algae were placed in 5 ml of reaction mixture, which contained 0.5 mM EDTA, 0.1% (v/v) propanol, 0.01 mM glucose and 50 mM $NaNO_3$ in 0.1 M KH_2PO_4 buffer (pH 8.0). The mixture was flushed with N_2 gas for 2 min, sealed immediately and incubated for 1 h in darkness so that the reaction occurred in the absence of oxygen and light. NR activity was determined by incubating 2 ml of this mixture with 1 ml 1% (w/v) of sulphanilamide and 1 ml of 0.1% (w/v) of N-(1-Naphthyl)ethylenediamine dihydrochloride at room temperature for 15 min. The amount of nitrite produced was determined by measuring absorbance at 540 nm. The NR activity was expressed as μmol $NO_2^{-1}\,g^{-1}$ FW h^{-1}.

Statistical analyses

All the data were expressed as means ± SD (n = 3). One-way ANOVA and the Tukey test were used to analyze differences among treatments with the software Origin 7.0 (Origin Lab Corp., Northampton, MA, USA). The Shapiro-Wilk and Levene tests for normality and variance checks were made prior to all parametric analyses. The significant p-value adopted was 0.05.

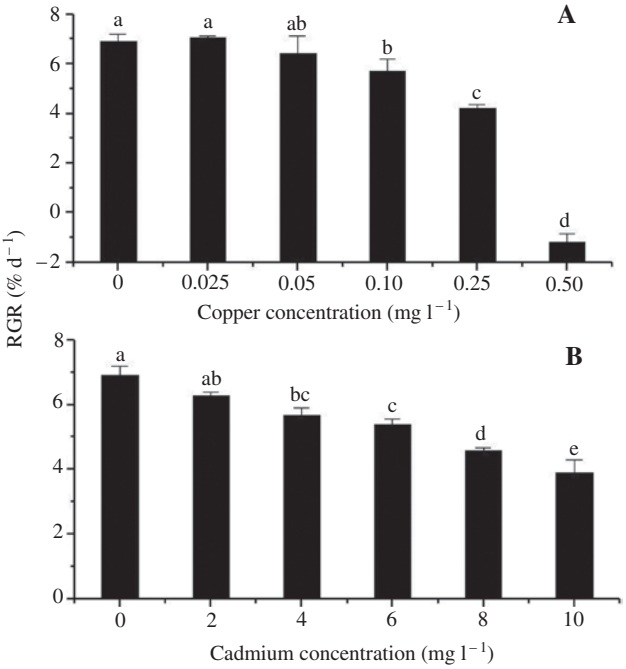


Figure 1: *Porphyra haitanensis*: Relative growth rate (RGR) of thalli exposed for 7 days to different concentrations of (A) copper and (B) cadmium.
Significant differences among the treatment means are indicated by different lowercase letters (Tukey test, $p < 0.05$). Values are means ± SD for triplicate samples.

Results

Growth

The mean RGR of *Porphyra haitanensis* was not significantly affected by 0.025 or 0.05 mg l^{-1} Cu but was decreased, relative to the control, at Cu concentrations of 0.10 mg l^{-1} and above and, at 0.50 mg l^{-1} Cu, RGR was negative (Figure 1A). Shrinkage and rot of the algae were observed at this concentration on the seventh day. Cd was less toxic than Cu. RGR was not significantly decreased at 2 mg l^{-1} Cd but exposure to 4, 6, 8 and 10 mg l^{-1} of Cd progressively reduced growth (Figure 1B).

Pigment content

Significant reductions in the content of both Chl *a* and Car were observed only when *Porphyra haitanensis* was exposed to the highest concentration of Cu investigated (0.50 mg l^{-1}; Figure 2A, C). For Cd, Chl *a* content declined only at 10 mg l^{-1} Cd, whereas Car content was reduced at both 8 and 10 mg l^{-1} Cd (Figure 2B, D).

The content of PE in *P. haitanensis* increased at 0.10 mg l^{-1} Cu and 4, 6 and 8 mg l^{-1} Cd (Figure 2E, F), but was lower than in the controls after exposure to 0.50 mg l^{-1} Cu and 10 mg l^{-1} Cd. Compared with the control, a significant decline in PC content was observed when *P. haitanensis* was exposed to 0.50 mg l^{-1} Cu or 10 mg l^{-1} Cd (Figure 2G, H).

Photosynthetic activity

Compared with the control, the F_v/F_m value of *Porphyra haitanensis* was progressively reduced at 0.10, 0.25 and 0.50 mg l^{-1} Cu and at 6, 8 and 10 mg l^{-1} Cd (Table 1). Similarly, P_m decreased progressively at Cu concentrations of 0.05 mg l^{-1} and above, and at Cd concentrations of 4 mg l^{-1} and above (Table 1). α, however, was inhibited only at 0.25 and 0.50 mg l^{-1} Cu and 10 mg l^{-1} Cd (Table 1), and dark respiration (R_d) was not significantly affected by any of the Cu or Cd treatments (Table 1).

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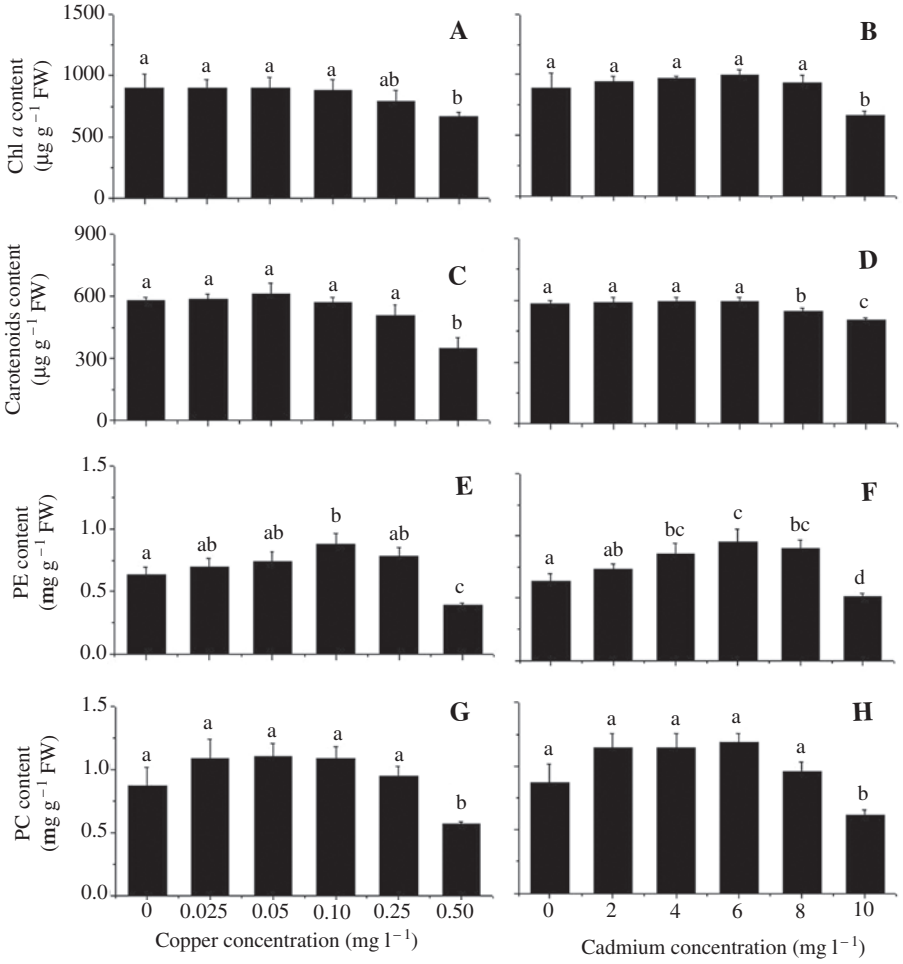


Figure 2: *Porphyra haitanensis*: Chlorophyll *a* (Chl *a*) (A, B), carotenoids (C, D), phycoerythrin (PE) (E, F) and phycocyanin (PC) (G, H) contents of thalli after exposure for 7 days to different concentrations of copper (A, C, E, G) and cadmium (B, D, F, H).
For other details, see Figure 1.

Table 1: *Porphyra haitanensis*: Effects of exposure to different copper and cadmium concentrations for 7 days on the optimal quantum yield (F_v/F_m) and photosynthetic parameters derived from photosynthesis vs. irradiance curves: P_m, maximal net photosynthetic rate (μmol O_2 g^{-1} FW h^{-1}); α, apparent photosynthetic efficiency (μmol O_2 g^{-1} FM h^{-1})/(μmol m^{-2} s^{-1}); R_d, dark respiration rate (μmol O_2 g^{-1} FW h^{-1}).

	F_v/F_m	P_m	α	R_d
Copper				
0 mg l^{-1}	0.613 ± 0.02^{a}	177.49 ± 3.75^{a}	0.94 ± 0.054^{a}	− 15.23 ± 1.92^{ab}
0.025 mg l^{-1}	0.619 ± 0.02^{a}	174.20 ± 4.65^{a}	0.93 ± 0.034^{a}	− 15.78 ± 1.12^{ab}
0.05 mg l^{-1}	0.591 ± 0.02^{ab}	158.85 ± 7.95^{b}	0.94 ± 0.033^{a}	− 17.81 ± 0.59^{a}
0.10 mg l^{-1}	0.539 ± 0.02^{b}	129.57 ± 4.29^{c}	0.89 ± 0.013^{ab}	− 17.42 ± 0.86^{ab}
0.25 mg l^{-1}	0.454 ± 0.03^{c}	115.86 ± 4.18^{d}	0.80 ± 0.073^{b}	− 17.57 ± 0.41^{ab}
0.50 mg l^{-1}	0.084 ± 0.02^{d}	51.64 ± 1.68^{e}	0.38 ± 0.023^{c}	− 14.92 ± 0.16^{b}
Cadmium				
0 mg l^{-1}	0.613 ± 0.02^{a}	177.49 ± 3.75^{a}	0.94 ± 0.054^{a}	− 15.23 ± 1.92^{a}
2 mg l^{-1}	0.607 ± 0.02^{a}	174.12 ± 3.28^{a}	0.94 ± 0.046^{a}	− 16.41 ± 1.63^{a}
4 mg l^{-1}	0.579 ± 0.02^{ab}	153.99 ± 4.42^{b}	0.94 ± 0.053^{a}	− 17.30 ± 0.88^{a}
6 mg l^{-1}	0.543 ± 0.02^{b}	139.95 ± 6.21^{c}	0.92 ± 0.059^{a}	− 16.86 ± 1.14^{a}
8 mg l^{-1}	0.473 ± 0.02^{c}	131.48 ± 4.09^{cd}	0.92 ± 0.038^{a}	− 17.56 ± 0.36^{a}
10 mg l^{-1}	0.404 ± 0.02^{d}	103.49 ± 6.84^{e}	0.73 ± 0.088^{b}	− 15.03 ± 1.33^{a}

Within each column of data, different superscript letters indicate significant differences at p = 0.05. Values are means ± SD (n = 3).

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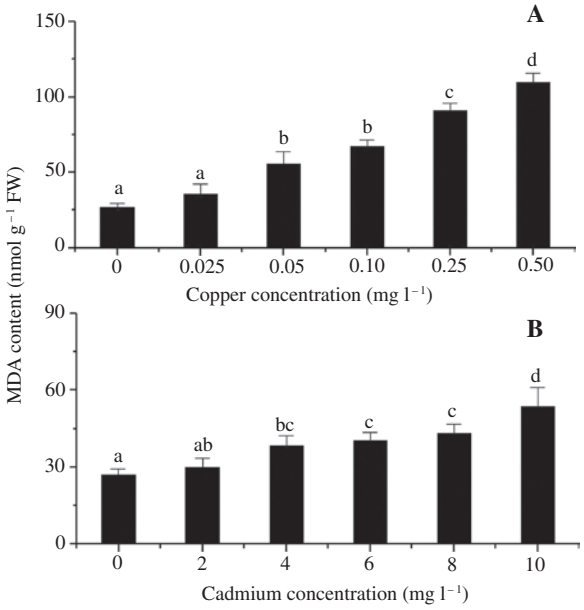


Figure 3: *Porphyra haitanensis*: Malondialdehyde (MDA) content of thalli after exposure for 7 days to different concentrations of (A) copper and (B) cadmium.
For other details, see Figure 1.

MDA content

The content of MDA increased with increasing concentrations of Cu or Cd (Figure 3A, B). Compared with the control, MDA content increased significantly at Cu concentrations of 0.05 mg l^{-1} and above or at Cd concentrations of 4 mg l^{-1} and above (Figure 3A, B).

Activities of antioxidant enzymes and nitrate reductase

A significant increase in SOD activity of *Porphyra haitanensis* was observed at 0.25 and 0.50 mg l^{-1} Cu or 8 mg l^{-1} Cd (Figure 4A, B), but the CAT activity was reduced by the highest concentration of Cu tested (0.50 mg l^{-1}; Figure 4C). In the Cd treatments, the CAT activity was increased significantly at 8 mg l^{-1} Cd but declined to the control values at 10 mg l^{-1} (Figure 4D). The activity of NR in *P. haitanensis* declined progressively with increasing Cu concentrations above 0.10 mg l^{-1} or Cd concentrations above 6 mg l^{-1} (Figure 4E, F).

Discussion

The present results showed that the physiological responses of *Porphyra haitanensis* to Cu were different from those to Cd. *P. haitanensis* exhibited a tolerance to low concentrations of both Cu and Cd, but Cu was more toxic than Cd within the ranges of concentrations used in the current study.

Cu and Cd are known to inhibit the normal growth of algae (Brown and Newman 2003, Xia et al. 2004, Zhu et al. 2011). The threshold concentration that caused a significant reduction in RGR for *P. haitanensis* was 0.10 mg l^{-1} Cu and 4 mg l^{-1} Cd. Li et al. (2010) reported that *P. haitanesis* was inhibited at 0.0635 mg l^{-1} Cu and above. Zhu et al. (2011) reported that the threshold concentrations for *Hizikia fusiformis* were 0.05 mg l^{-1} Cu and 4 mg l^{-1} Cd. *P. haitanensis* is, therefore, more tolerant of Cu stress than *H. fusiformis*, whereas the toxicity of Cd was the same for the two species. Our results indicate that Cu is more toxic to *P. haitanensis* than Cd, as also reported for *Gracilaria lemaneiformis* and *H. fusiformis* (Xia et al. 2004, Zhu et al. 2011). Such different toxic effects of Cu and Cd on macroalgae may reflect differential tolerance within species (Baumann et al. 2009, Zhu et al. 2011). The toxicity and uptake mechanisms for the two metals may be different and different macroalgae may have a resistance mechanism or an exclusion mechanism which reduces metal uptake into the cells (Baumann et al. 2009, Babu et al. 2014). Cu is considered to be the most toxic metal to algae after mercury (Bertrand and Poirier 2005). The redox-active nature of Cu causing the formation of highly reactive $^{\cdot}OH$ through cycling between Cu^{+} and Cu^{2+} is one of the most important mechanisms that can reduce growth rate (Halliwell and Gutteridge 1999, Collén et al. 2003). Cu-induced interference with cell division and/or expansion is a possible reason for the observed reduction in growth (Stauber and Florence 1987). This may be linked to a decrease in turgor and/or a change in cell wall elasticity due to Cu toxicity (Brown and Newman 2003). In contrast to Cu, Cd has no known metabolic function in algae and cannot contribute to $^{\cdot}OH$ formation in the Fenton reaction (Halliwell and Gutteridge 1999), but it can be taken up by algae and cause disturbances in photosynthesis, resulting in reduction in RGR (Collén et al. 2003, Zhu et al. 2011). *P. haitanensis* may actively exclude Cd to minimize the metal's toxic impact and the lower solubility of Cd can result in greater chelation and adsorption on particles in the culture medium (Zhu et al. 2011). These factors can also explain the greater toxic effect of Cu than of Cd to *P. haitanensis*.

Chl *a*, Car and phycobiliproteins are the major photosynthetic pigments of red macroalgae and they are key targets for metals, frequently used to assess metal stress (Xia et al. 2004, Zhu et al. 2011). In the present study, the Chl *a* and Car contents of *P. haitanensis* declined

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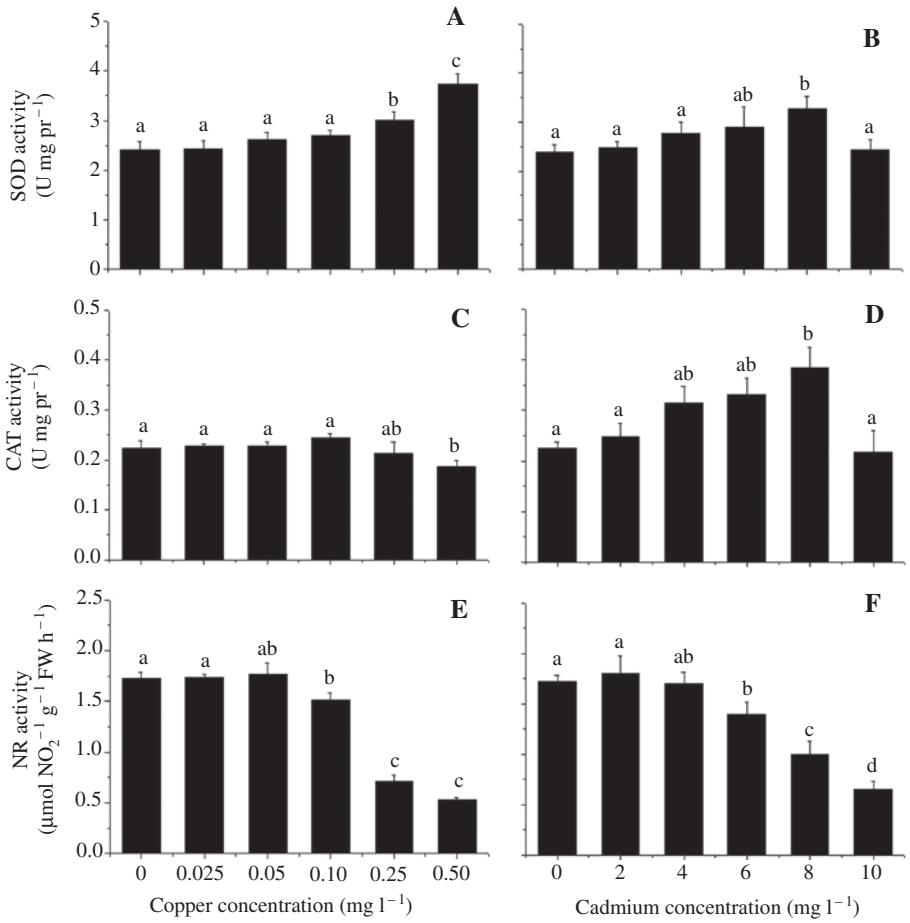


Figure 4: *Porphyra haitanensis*: Superoxide dismutase (SOD) (A, B), catalase (CAT) (C, D), nitrate reductase (NR) (E, F) activities of thalli after exposure for 7 days to different concentrations of copper (A, C, E) and cadmium (B, D, F).
For other details, see Figure 1.

significantly at high Cu or Cd concentrations. The direct peroxidative breakdown of pigments and chloroplast membrane lipids by the ROS induced by Cu stress is one of the most important reasons for the decrease of Chl *a* and Car contents (Sandamann and Böer 1980). The reduced chlorophyll content in Cd-treated samples has been found to be associated with magnesium (Mg) and iron (Fe) deficiency in the process of chlorophyll biosynthesis (Greger and Ögren 1991). Cu and Cd can destroy the synthetic pathway for carotene and lutein and this may be the reason for the decrease in Car content of *P. haitanensis* in this study. In contrast to Chl *a* and Car, PE and PC increased at low concentrations of Cu and Cd and decreased at the highest Cu and Cd concentrations. Therefore, PE and PC were more sensitive to Cu or Cd stress. This phenomenon has also been observed in *G. lemaneiformis* (Xia et al. 2004). Moreover, PE was more sensitive than PC to Cu or Cd treatments in this study. In addition, we found that the toxic effect of Cu on PE and PC in *P. haitanensis* was greater than that of Cd. It may be that Cu has a greater binding affinity for PE and PC, and thus can cause serious damage to these pigments (Brown and Newman 2003). The increasing contents of PE and PC may serve as an efficient protective mechanism that can accelerate photosynthesis. The reason for the decrease of PE and PC contents is that the phycobilisomes could have been damaged in high Cu and Cd concentrations (Xia et al. 2004). Decreases in the content of photosynthetic pigments is, therefore, an important toxic effect underlying Cu or Cd stress, which can further affect the photosynthesis and growth of algae.

Chlorophyll fluorescence analysis using PAM techniques has been used to monitor the toxic effects of metals on algae (Baumann et al. 2009, Zhu et al. 2015). The chlorophyll fluorescence parameter F_v/F_m is proportional to the quantum yield of photochemistry and represents the efficiency of the photosynthetic apparatus of PS II in macroalgae (Baumann et al. 2009). In the present study, F_v/F_m decreased significantly at Cu concentrations of

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0.10 mg l^{-1} and above, or at Cd concentrations of 6 mg l^{-1} and above. This coincided with the sharp decline of RGR and implied that *P. haitanensis* was under serious stress. The inhibition of electron donation to the primary photochemical reaction is considered to be the most important reason for the Cu-induced decline in F_v/F_m (Brown and Newman 2003). In contrast, the decline of F_v/F_m in response to Cd may be caused by the substitution of the Mg^{2+} ion at the centre of the chlorophyll molecule by Cd^{2+}, a process that lowers the fluorescence quantum yield and reduces the activity of PS II (Küpper et al. 1996). In addition, the decrease of photosynthetic pigments at high Cu or Cd concentrations may be an underlying effect. Our results showed that the toxic effect of Cu on F_v/F_m in *P. haitanensis* was greater than that of Cd.

The toxic effects of metals on photosynthesis are mainly related to the metal-induced structural and functional modifications of the photosynthetic machinery that have been reported in some algae (Xia et al. 2004, Aggarwal et al. 2011, Zhu et al. 2011, 2015). In this study, P_m decreased with increasing Cu or Cd concentrations, indicating that photosynthesis was inhibited and that the light-harvesting complex and the oxygen-evolving complex may have been impaired. A significant decline of α at high Cu or Cd concentrations suggested a reduction in the efficiency of light harvesting and energy conversion. The reason for the decline in photosynthesis of *P. haitanensis* may be that the excess Cu damages the proteins of the oxygen-evolving complex at the donor site of PS II (Pätsikkä et al. 2001). In contrast, the Cd-induced decline in photosynthesis may be because Cd can damage the photosynthetic apparatus (Siedlecka and Krupa 1996). In addition, the reduction of Chl *a* and Car contents and the phycobilisome damage under high Cu and Cd concentrations would result in a decline of light energy absorbed. Our results also indicate that the greater toxicity of Cu than of Cd on photosynthesis in *P. haitanensis* coincides with their effects on growth, pigments and F_v/F_m. Such a phenomenon has been reported previously in *G. lemaneiformis* and *H. fusiformis* (Xia et al. 2004, Zhu et al. 2011). The reason may be that the redox-active nature of Cu can cause the formation of highly reactive $^{\cdot}OH$, resulting in conformational changes to enzymes and proteins, thereby preventing their normal functions and inhibiting the photosynthesis and growth of *P. haitanensis*.

Metals can induce oxidative stress by generating ROS, leading to lipid peroxidation and increasing MDA content in algae (Collén et al. 2003, Zhu et al. 2011, 2015). MDA is an indicator of lipid peroxidation induced by metals (Zhu et al. 2015). In this study, a significant increase in the content of MDA in *P. haitanensis* was observed at Cu concentrations of 0.05 mg l^{-1} and above, or at Cd concentrations of 4 mg l^{-1} and above. This suggests that the two metals can generate an oxidative stress in *P. haitanensis*. Our results also showed that Cu induced greater accumulation of MDA than Cd. This agrees with our previous findings in *H. fusiformis* (Zhu et al. 2011). It appears that the redox-active nature of Cu can catalyse the formation of reactive $^{\cdot}OH$, inducing more oxidative stress and MDA formation than Cd. As Cd is a non-redox metal, its ability to induce oxidative stress and increase MDA content in *P. haitanensis* is most likely an indirect effect (Collén et al. 2003, Zhu et al. 2011).

Exposure to metals can induce the production of ROS, such as $O_2^{\cdot-}$, $^{\cdot}OH$ and H_2O_2, directly or indirectly, and algae can resist the damage caused by ROS through protective mechanisms, such as those involving SOD, CAT, glutathione and Car (Okamoto et al. 2001, Collén et al. 2003, Zhu et al. 2011). The metalloenzyme SOD is the major $O_2^{\cdot-}$ scavenger and CAT can eliminate H_2O_2 in algae (Collén et al. 2003). In this study, the SOD activity of *P. haitanensis* increased significantly at 0.10 and 0.50 mg l^{-1} Cu or 8 mg l^{-1} Cd. This suggests that $O_2^{\cdot-}$ accumulated with increasing Cu and Cd concentrations. The activity of CAT was reduced at 0.50 mg l^{-1} Cu but increased at 8 mg l^{-1} Cd. The increase in activity of CAT may reduce the concentration of H_2O_2 and buffer the oxidative stress, but the decrease may be due to the $O_2^{\cdot-}$ radicals produced under Cu stress, as CAT activity can be inhibited by increased levels of $O_2^{\cdot-}$ (Cakmak 2000, Bertrand and Poirier 2005). How other enzymatic and non-enzymatic defense systems respond to Cu and Cd stress should be studied in the future.

NR is one of the most important enzymes of any biosynthetic pathway in the algae (Berges 1997). The NR activity in macroalgae is influenced by many environmental factors such as light, CO_2 concentration and metals (Chow et al. 2004, Zou 2005, Han et al. 2008, Zhu et al. 2011, 2015). In this study, the NR activity of *P. haitanensis* declined progressively with increasing Cu concentrations above 0.10 mg l^{-1} or Cd concentrations above 6 mg l^{-1}. Our results indicate that the elevated levels of oxidative stress induced by Cu and Cd can impair the NR activity and further disturb N_2 metabolism and affect the growth of algae (Zhu et al. 2011).

In conclusion, *P. haitanensis* was capable of tolerating Cu and Cd stress at low concentrations. However, *P. haitanensis* was negatively impacted at high Cu and Cd concentrations. Cu exhibited greater toxic effects on *P. haitanensis* than Cd. This may be related to the redox-active nature of Cu. Our results provide additional evidence that PE and PC are more sensitive to Cu and Cd than Chl *a* and Car. The toxic effects of these metals on

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the physiology of *P. haitanensis* include reduction of photosynthetic pigment content, depression of chlorophyll fluorescence and photosynthesis rate, induction of lipid oxidation, changes in the activities of SOD, CAT and NR and inhibition of growth. *Porphyra haitanensis* appears to be able to resist the toxic effects of Cu and Cd through protective mechanisms, such as increasing the content of PE and the activities of SOD, CAT and other antioxidant enzymes. *Porphyra haitanensis* can be used as a bioindicator of metal pollution in the future. In addition, we should focus on the nutritional value of this species under increasing metal pollution, especially metal toxicity.

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Bionotes



Xifeng Zhu
The Key Laboratory of Animal Nutrition and Feed Science (South China) of Ministry of Agriculture, Guangdong Open Laboratory of Animal Breeding and Nutrition, Guangdong Key Laboratory of Animal Breeding and Nutrition, Institute of Animal Science, Guangdong Academy of Agricultural Sciences, Guangzhou, Guangdong 510640, China
xfzhu1983@163.com

Xifeng Zhu is Assistant Professor of Fisheries at the Institute of Animal Science, Guangdong Academy of Agricultural Science. He was awarded an MS in Marine Biology by Shantou University. More recent research has concentrated on algal physiology and animal nutrition and feed science.

Dinghui Zou
College of Environment and Energy, South China University of Technology, Guangzhou 510006, China; and The Key Laboratory of Pollution Control and Ecosystem Restoration in Industry Clusters, Ministry of Education, China

Dinghui Zou is Professor of Marine Biology at the College of Environment and Energy, South China University of Technology. He was awarded a PhD in Hydrobiology by the Institute of Hydrobiology, Chinese Academy of Sciences. More recent research has concentrated on algal physiology and applied phycology.

Yanhua Huang
The Key Laboratory of Animal Nutrition and Feed Science (South China) of Ministry of Agriculture, Guangdong Open Laboratory of Animal Breeding and Nutrition, Guangdong Key Laboratory of Animal Breeding and Nutrition, Institute of Animal Science, Guangdong Academy of Agricultural Sciences, Guangzhou, Guangdong 510640, China

Yanhua Huang is Professor of Animal Nutrition and Feed Science at the Institute of Animal Science, Guangdong Academy of Agricultural Science. She was awarded a PhD in Animal Nutrition and Feed Science by South China Agricultural University. More recent research has concentrated on animal nutrition and feed science.

Junming Cao
The Key Laboratory of Animal Nutrition and Feed Science (South China) of Ministry of Agriculture, Guangdong Open Laboratory of Animal Breeding and Nutrition, Guangdong Key Laboratory of Animal Breeding and Nutrition, Institute of Animal Science, Guangdong Academy of Agricultural Sciences, Guangzhou, Guangdong 510640, China

Junming Cao is Professor of Animal Nutrition and Feed Science at the Institute of Animal Science, Guangdong Academy of Agricultural Science. He was awarded a PhD in Nutritional Physiology and Biochemistry by the University of Burgundy. More recent research has concentrated on animal nutrition and feed science.

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Yuping Sun
The Key Laboratory of Animal Nutrition and Feed Science (South China) of Ministry of Agriculture, Guangdong Open Laboratory of Animal Breeding and Nutrition, Guangdong Key Laboratory of Animal Breeding and Nutrition, Institute of Animal Science, Guangdong Academy of Agricultural Sciences, Guangzhou, Guangdong 510640, China

Yuping Sun is Associate Professor of Animal Nutrition and Feed Science at the Institute of Animal Science, Guangdong Academy of Agricultural Science. She was awarded a PhD in Hydrobiology by Jinan University. More recent research has concentrated on animal nutrition and aquaculture.

Bing Chen
The Key Laboratory of Animal Nutrition and Feed Science (South China) of Ministry of Agriculture, Guangdong Open Laboratory of Animal Breeding and Nutrition, Guangdong Key Laboratory of Animal Breeding and Nutrition, Institute of Animal Science, Guangdong Academy of Agricultural Sciences, Guangzhou, Guangdong 510640, China

Bing Chen is Associate Professor of Fisheries at the Institute of Animal Science, Guangdong Academy of Agricultural Science. She was awarded an MS in Aquaculture by South China Agricultural University. More recent research has concentrated on animal nutrition and aquaculture.

Xiaoying Chen
The Key Laboratory of Animal Nutrition and Feed Science (South China) of Ministry of Agriculture, Guangdong Open Laboratory of Animal Breeding and Nutrition, Guangdong Key Laboratory of Animal Breeding and Nutrition, Institute of Animal Science, Guangdong Academy of Agricultural Sciences, Guangzhou, Guangdong 510640, China

Xiaoying Chen is Assistant Professor of Fisheries at the Institute of Animal Science, Guangdong Academy of Agricultural Science. She was awarded an MS in Aquaculture by South China Agricultural University. More recent research has concentrated on animal nutrition and aquaculture.

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